Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing (with Business Management)**

Assessment	**Computer Systems 2.0**
Assessment Type	**Global Examination**
Date	**Wednesday 5th March 2025**
Time	**2.00PM-5.00PM**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions
Do NOT copy the questions to your answer booklet

Marks

Question 1
For each answer, only provide the corresponding letter (e.g., i, ii, iii, or iv).

a) Which of the following is NOT a characteristic of modern computer hardware? **1**

i) CPU ii) Memory

iii) Operating System iv) Cache

Mark Scheme:
iii) Operating System (1 mark)

b) Which component of a CPU is responsible for directing the flow of data and instructions? **1**

i) Control Unit ii) Memory

iii) ALU iv) Cache

Mark Scheme:
i) Control Unit (1 mark)

c) Which of the following is a characteristic of quantum computing? **1**

i) Uses classical bits for processing ii) Utilises qubits to represent and process information

iii) Operates on a binary logic framework iv) Relies solely on traditional hardware components

Mark Scheme:
ii) Utilises qubits to represent and process information (1 mark)

d) Which of the following is used to connect a CPU to other components in a computer? **1**

i) Cache ii) Memory

iii) Bus iv) Control Unit

Mark Scheme:
iii) Bus (1 mark)

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Marks

e) In which numbering system is the digit 'F' used to represent a value? **1**

i) Hexadecimal

ii) ASCII

iii) Octal

iv) Extended ASCII

Mark Scheme:
i) Hexadecimal (1 mark)

f) Which protocol is responsible for ensuring error-free transmission in data networks? **1**

i) HTTP

ii) SMTP

iii) FTP

iv) TCP

Mark Scheme:
iv) ***TCP (1 mark)***

g) What is the primary challenge for quantum computing today? **1**

i) Lack of enough qubits to perform complex tasks

ii) High power consumption

iii) Difficulty in interfacing with classical systems

iv) Limited quantum algorithms available

Mark Scheme:
i) Lack of enough qubits to perform complex tasks (1 mark)

h) The primary function of memory in a computer system is to: **1**

i) Process data

ii) Communicate with external devices

iii) Control data flow

iv) Store data temporarily or permanently

Mark Scheme:
iv) Store data temporarily or permanently (1 mark)

i) Which of the following is **TRUE** about a computer bus? **1**

i) It is used only for transferring data between memory and the CPU

ii) It connects the CPU to external devices like printers and monitors

iii) It is a collection of wires that transfers data, addresses and control signals between various components of a computer system

iv) It can only transfer data in one direction at a time

Mark Scheme:
iii) It is a collection of wires that transfers data, addresses and control signals between various components of a computer system (1 mark)

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Marks

j) Which of the following is a core characteristic of a distributed system? **1**

i) Centralised control

ii) A single computer manages all tasks

iii) Multiple computers work together as a single system

iv) All resources located in a single physical location

Mark Scheme:
iii) Multiple computers work together as a single system (1 mark)

Total 10 Marks

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Marks

Question 2

a) Outline TWO (2) tasks carried out by the OS. Provide a description for each. **6**

Mark Scheme:

1 mark per operating system purpose outlined to a maximum of 2 marks.

Up to 2 marks for additional valid points per outlined purpose to a maximum of 4 marks.

Note: a description does not need to be exhaustive (the series of accepted answers are distinguished by OR / AND / LIKE / WHILE)

<u>Any 2 from:</u>

- ***Resource Management (1 mark) – Allocates and manages hardware resources (1 mark) LIKE CPU OR memory OR storage devices OR peripherals (1 mark)***
- ***Process Management (1 mark) - Oversees process execution OR scheduling tasks OR managing process states (1 mark) AND facilitating inter-process communication and synchronisation (1 mark)***
- ***Memory Management (1 mark) - Controls memory allocation to processes, manages virtual memory (1 mark) AND optimises memory usage by swapping data between RAM and secondary storage devices (1 mark)***
- ***File System Management (1 mark) - Organises and manages files and directories allowing users to create OR read OR write OR delete files (1 mark) WHILE ensuring data integrity and security through permissions and access control. (1 mark)***
- ***Device Management (1 mark) - Interacts with hardware devices, providing device drivers for software applications to communicate with peripherals (1 mark) LIKE keyboards, mice, printers, and network adapters. (1 mark)***
- ***Security (1 mark) - Implements security measures like user authentication OR access control OR encryption OR firewall configurations (1 mark) to protect the system from unauthorised access OR malware OR safeguarding sensitive data and resources. (1 mark)***

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Marks

b) Describe the difference between pre-emptive and non-pre-emptive scheduling in an operating system. **2**

Mark Scheme:

Pre-emptive scheduling allows interruptions during process execution. (1 mark)

Non-pre-emptive scheduling does not allow interruptions. (1 mark)

c) Other than fault tolerance and reliability, state and describe ONE (1) additional distinct characteristic of a RTOS's operation concerning task and resource management. **2**

Mark Scheme:

<u>Any 1 example and accompanying description from below:</u>

- ***Pre-emption (1 mark): An RTOS allows higher-priority tasks to pre-empt lower-priority tasks for timely execution. (1 mark)***
- ***Fixed or Dynamic Task Allocation: Tasks may be statically or dynamically assigned to processors based on workload or priority. (1 mark)***
- ***Resource Allocation (1 mark): The RTOS efficiently allocates system resources like CPU, memory, and peripherals among tasks. (1 mark)***
- ***Memory Management: Simplified memory management techniques, such as fixed-size memory pools, ensure predictable behaviour. (1 mark)***
- ***Real-Time Clock and Timers (1 mark): High-resolution timers and real-time clocks are used for precise scheduling and time-sensitive task execution. (1 mark)***
- ***Task Creation and Termination (1 mark): The RTOS supports dynamic creation, deletion, and management of tasks based on system needs. (1 mark)***
- ***Multicore and Multithreading Support (1 mark): The RTOS efficiently distributes tasks across multiple cores or threads to meet real-time requirements. (1 mark)***
- ***Task States and Context Switching (1 mark): The RTOS manages task states and performs efficient context switching to ensure optimal task execution. (1 mark)***

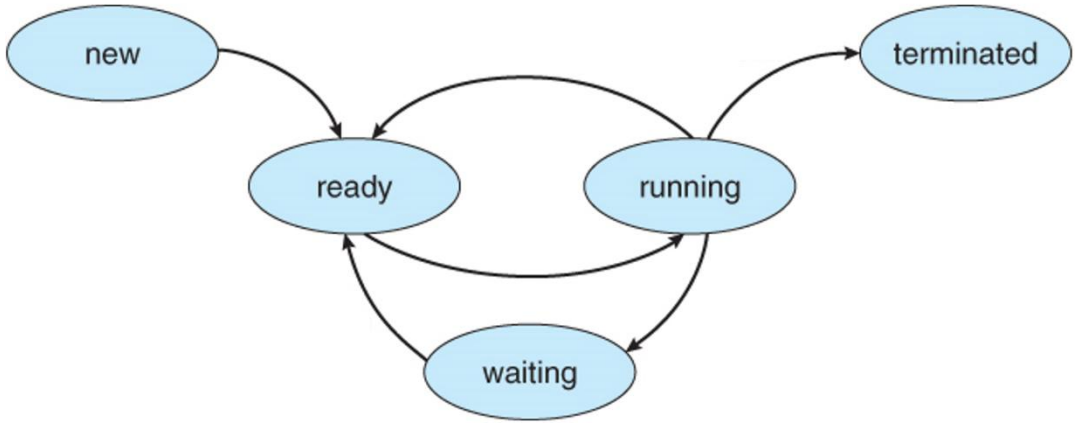
Total 10 Marks

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Marks

Question 3

a) Using the following diagram, explain each of the FIVE (5) different states a process can be in. **5**



Mark Scheme:

<u>States of the CPU:</u>

- ***New - A process is being created. (1 mark)***
- ***Ready - The process is ready to execute but is waiting for CPU time. (1 mark)***
- ***Running - The CPU executes the process. (1 mark)***
- ***Waiting (Blocked) - A process is unable to continue because it is waiting for some external event (e.g., I/O completion or resource availability). (1 mark)***
- ***Terminated (Exit) - The process has completed execution or has been terminated due to an error. (1 mark)***

Accept any other reasonable correct answers.

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Marks

b) Timesharing is a technique associated with the management of a multi-user system. **4**

Describe TWO (2) distinct advantages and TWO (2) distinct disadvantages of timesharing.

Mark Scheme:

<u>Stated Advantages and Disadvantages must be inherently different from one another</u>.

Advantages (Any 2 from):

- ***Resource Efficiency: Timesharing maximizes the use of CPU and memory resources. (1 mark)***
- ***Perceived Fairness: A timeshared system works by allocating each user a measurable time slot that is equal for every user. (1 mark)***
- ***Cost-Effective: Timesharing systems can support multiple users on a single machine. (1 mark)***
- ***Supports Multitasking: Allows multiple applications to run concurrently. (1 mark)***
- ***Flexibility: Timesharing systems can adjust the allocation of resources dynamically, helping to balance loads based on demand. (1 mark)***

Disadvantages (Any 2 from):

- ***Reduced Performance Under High Load: As more users or processes are added, each one gets a smaller slice of time, which can reduce performance. (1 mark)***
- ***Complexity of Management: Requires sophisticated scheduling and resource management to ensure fair distribution and prevent issues, such as deadlock. (1 mark)***
- ***Security and Privacy Risks: Security concerns if isolation between user sessions is not properly maintained. (1 mark)***
- ***Dependency on Stable Software: Timesharing requires reliable system software to handle the frequent switching of processes. (1 mark)***

Accept any other reasonable correct answers.

c) State where both program data and instructions are stored in a Von Neumann architecture. **1**

Mark Scheme:

In the main memory (1 mark)

Total 10 Marks

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Marks

Question 4

a) Convert the following decimal value to binary: **87**. Provide your answer in 8 bits. **3**

Show your workings.

Mark Scheme:

87 ÷ 2 = 43 remainder 1
43 ÷ 2 = 21 remainder 1
21 ÷ 2 = 10 remainder 1
10 ÷ 2 = 5 remainder 0
5 ÷ 2 = 2 remainder 1
2 ÷ 2 = 1 remainder 0
1 ÷ 2 = 0 remainder 1
Final answer: 01010111

1 mark for displaying workings (workings may differ from format above)
1 mark for correct answer
1 mark for correct answer in 8 bits (additional 0 added to left side)

b) Subtract hexadecimal 58 from A2 in hexadecimal. Express the result in hexadecimal. **4**

Show your working.

Mark Scheme:

A = 10 in decimal, A2 = (10 × 16) + 2 = 160 + 2 = 162 (Decimal)

58 = (5 × 16) + 8 = 80 + 8 = 88 (Decimal)

162 – 88 = 74 (Decimal)

74 ÷ 16 = 4 remainder 10, 10 (Decimal) = A (hex) , so the result is 4A.

1 mark for correct conversion of A2
1 mark for correct conversion of 58
1 mark for correct value from subtraction calculation
1 mark for correct answer displayed in hexadecimal format.

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Marks

c) Add the unsigned binary numbers 10101 and 00100. Give the result in binary. **3**

Show your workings.

Mark Scheme:

	16	**8**	**4**	**2**	**1**
	1	0	1	0	1
+	0	0	1	0	0
	1	**1**	**0**	**0**	**1**
		1			

1 mark for displaying workings
1 mark for evidence of carry
1 mark for correct answer

Total 10 Marks

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Marks

Question 5

a) Name the THREE (3) main stages of a CPU's instruction cycle. **3**

Mark Scheme:

Fetch instruction from RAM (1 mark)
Decode instruction (1 mark)
Execute instruction (1 mark)

b) Explain the purpose of an operand in a CPU instruction? **2**

Mark Scheme:

To represent the data on which an operation is to be performed. (1 mark) It holds the data needed for executing instructions such as arithmetic, logic and data manipulation operations. (1 mark)

c) Explain the purpose of an opcode in a CPU instruction. Give ONE (1) example. **3**

Mark Scheme:

To specify the operation that the CPU should perform. (1 mark) It serves as a code that indicates the type of operation, such as addition OR subtraction OR data movement (1 mark), enabling the CPU to execute the appropriate instruction (1 mark).

d) Before the CPU processes any data, it is recommended that data is checked via a parity checker. Describe how a parity checker works. **2**

Mark Scheme:

A parity checker for a CPU is a component or circuitry responsible for verifying the integrity of data transmitted to or stored in the CPU's memory (1 mark). It works by adding an extra parity bit to each byte of data, which is calculated based on the number of bits set to 1 in the byte (either even or odd parity). (1 mark)

Total 10 Marks

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Marks

Question 6

a) Explain FOUR (4) requirements to ensure successful global network communication. **4**

Mark Scheme:

Interoperability: across multi-platforms (1 mark)
Flexibility: mixing and matching devices (1 mark)
Transparency: aids development and testing (1 mark)
Scalability: room for large scale growth 1 (mark)

Answers may differ from the above, but if the fundamental points are valid accept variants.

b) State the difference between HTTP and HTTPS. **1**

Mark Scheme:

HTTPS is encrypted and therefore more secure than HTTP. (1 mark)

Accept encrypted OR more secure.

c) In the OSI 7-layer reference model of network communications, what layer does logical addressing take place within? **1**

Mark Scheme:

Layer 3 OR networking layer (1 mark)

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Marks

d) Bus and Star represent TWO (2) different types of network configurations. **4**

For each network, state their fault tolerant and performance characteristics.

Mark Scheme:

Bus Topology: Fault Tolerance

- ***Single point of failure: if backbone cable fails, network is disrupted. (1 mark)***

Bus Topology: Performance

- ***Network performance deteriorates as more devices are added due to shared bus. (1 mark)***

Star Topology: Fault Tolerance

- ***Single Point of Failure at the Hub: If the central hub or switch fails, the entire network is affected. (1 mark)***

Star Topology: Performance

- ***Performance is generally stable, unless so many devices are added that overwhelm the central hub. (1 mark)***

Answers may differ from the above, but if the fundamental points are valid accept variants.

Total 10 Marks

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Marks

Question 7

a) A logic gate has TWO (2) inputs and ONE (1) output. The only time its output is TRUE is when both inputs are TRUE. Identify the type of this gate. **2**

What gate will it be called if both inputs are negated?

Mark Scheme:

AND gate (1 mark), NAND gate (1 mark)

b) An electrical system has TWO (2) inputs that control the flow of power. Power stops only when both inputs are closed. What logic does this represent? **1**

Mark Scheme:

OR logic (1 mark)

c) What logic name is given to the system where electricity stops only when both inputs are open? **1**

Mark Scheme:

NOR logic (1 mark)

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Marks

d) Complete the following table for the logic function: **A.(B+C)** **6**

A	***B***	***C***	B+C	A.(B+C)
0	*0*	*0*	0	0
0	*0*	*1*	1	
0	*1*	*0*		0
0	*1*	*1*	1	
1	*0*	*0*	0	0
1	*0*	*1*	1	
1	*1*	*0*	1	1
1	*1*	*1*		

Mark Scheme:

1 mark per correct table cell.

A	***B***	***C***	***B OR C / (B+C)***	***A AND (B OR C) / (A.(B+C)***
0	*0*	*0*	*0*	*0*
0	*0*	*1*	*1*	*0*
0	*1*	*0*	*1*	*0*
0	*1*	*1*	*1*	*0*
1	*0*	*0*	*0*	*0*
1	*0*	*1*	*1*	*1*
1	*1*	*0*	*1*	*1*
1	*1*	*1*	*1*	*1*

Total 10 Marks

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Marks

Question 8

a) Explain how system software differs from application software. **3**

Mark Scheme:

1 mark for each of the following, a maximum of 3 marks

- ***System software is used to support applications.***
- ***System software has higher privileges than applications.***
- ***Some system software can interface with hardware.***
- ***Application software is designed to perform specific tasks for users, like word processing or playing games,***
- ***system software manages the computer's hardware***

b) An example of system software that is essential for producing application software is the language compiler. Explain the main function of a compiler. **2**

Mark Scheme:

The compiler takes high-level language statements (1 mark) and converts them to executable low-level code, i.e. binary instructions. (1 mark)

c) When developers create software, they sometimes use virtual machines. Explain what a virtual machine is and what it is used for. **5**

Mark Scheme:

A virtual machine emulates OR simulates (1 mark) a guest computer's instructions on a host computer (1 mark).

It is used to evaluate other operating systems on a host operating system (1 mark); it is used to run applications written for one guest operating system on a host operating system (1 mark); it is used to run multiple guest operating systems on host operating system (1 mark).

Total 10 Marks

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Marks

Question 9

a) There are TWO (2) types of instruction set computers. Provide the full name of each type and describe TWO (2) distinct characteristics of each type. **6**

Mark Scheme:

1 mark for full name of each instruction set computer with a maximum of 2 marks for each characteristic for each type.

Complex Instruction Set Computers (CISC) – 1 mark
- ***The instructions do complex operations created to suit high-level language statements (1 mark)***
- ***Complex instructions require many CPU clock cycles (1 mark)***
- ***Take up less memory space (1 mark)***
- ***Less instructions required in programs (1 mark)***

Reduced Instruction Set Computers (RISC) – 1 mark
- ***Use simpler instructions (1 mark)***
- ***Simpler instructions require fewer CPU clock cycles (1 mark)***
- ***Take up larger memory space (1 mark)***
- ***More instructions required in programs (1 mark)***

b) Context switching is required when program control is re-directed to a different address location temporarily. What might cause this re-direction? **1**

Mark Scheme:

Any 1 from the below up to 1 mark:

An external interrupt. (1 mark)
Time sharing time-out (1 mark)

Accept other variants e.g., keyboard, mice, touchpad input (as examples of external interrupts)

c) When context switching takes place, the current program context must be saved. Describe what is meant by the program context. **3**

Mark Scheme:

PC register content (next instruction address) – (1 mark)
Status register content – (1 mark)
Additional register value(s) – (1 mark)

Total 10 Marks

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Marks

Question 10

a) Briefly describe ONE (1) advantage of a multi-core **and** ONE (1) advantage of a quantum computing system. **3**

State which of these systems might be better suited for running accounting software.

Mark Scheme:

Multi-core Systems: Parallel processing OR Energy Efficiency OR Improved Multi-tasking (1 mark).

Quantum computing Systems: Exponential Processing Power OR Solving Complex Problem (1 mark).

A multi-core system would be best utilised for running accounting software (1 mark).

b) When creating software, programmers must be considerate of utilising system resources efficiently. In what TWO (2) ways can the compiler help with optimisation of code? **2**

Mark Scheme:

Any 2 from:

A compiler helps to remove redundant code (1 mark)
Constant folding and propagation (1 mark)
Loop optimisation (1 mark)

c) Describe TWO (2) benefits of using distributed computer systems. **2**

Mark Scheme:

Any 2 from:

- ***Resource Sharing (both hardware and software) (1 mark)***
- ***Increased Computational Power (1 mark)***
- ***Remote process management (1 mark)***
- ***Increased Reliability (or Fault Tolerance) (1 mark)***
- ***Scalability (or Incremental Growth) (1 mark)***
- ***Multiple platform interoperability (1 mark)***

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		Marks
d)	Explain THREE (3) key characteristics of embedded systems and how they impact their design and performance.	**3**

Mark Scheme:

- ***Real-time Operation*** - ***Embedded systems often require real-time responses, demanding careful timing and efficient task scheduling to meet deadlines. (1 mark)***
- ***Resource Constraints - Limited processing power, memory, and storage require optimized code and efficient resource management. (1 mark)***
- ***Specialised Functionality - Designed for specific tasks, enabling tight hardware-software integration for better performance and reliability. (1 mark)***

Total 10 Marks

End of paper

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Learning Outcomes Matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	a-LO1, b-LO1, c-LO6, d-LO1, e-LO3, f-LO7, g-LO6, h-LO1, i-LO1, j-LO6	Yes
2	a-LO2, b-LO2, c-LO1/6	Yes
3	a-LO2, b-LO2, c-LO1	Yes
4	a/b/c-LO3	Yes
5	a/b/c/d-LO3	Yes
6	a/b/c-LO7	Yes
7	a/b/c/d-LO3	Yes
8	a-LO5, b-LO5, c-LO2/5	Yes
9	a/b/c-LO4	Yes
10	a-LO6, b-LO6, c-6, d-1/6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand and identify the main types and components of computer systems	None or inadequate description of computer systems and components	Some credible but incomplete attempt to demonstrate understanding of computer systems and components	Limited but satisfactory demonstration of understanding of computer systems and components	Good demonstration of understanding of computer systems and components	Excellent demonstration of understanding of computer systems and components
Describe the structure and role of modern operating systems	None or very basic description of components of operating systems	A very limited and basic description of components of operating systems	Limited and above basic description of components of operating systems	Good description of most components of operating systems	Excellent and full description of all components of operating systems
Understand and work with binary numbers and computer logic	No demonstration of working with binary numbers and logic operators	A basic demonstration of restricted range of working with binary numbers and logic operators	An above basic demonstration of working with restricted range of binary numbers and logic operators	A good demonstration of working with a wide range of binary numbers and logic operators	An excellent demonstration of working with full range of binary numbers and logic operators
Understand the essential structure of computer programs	None or trivial explanation of instruction sets, addressing modes and interrupts	Limited understanding and explanation of instruction sets, addressing modes and interrupts	Basic understanding and explanation of instruction sets, addressing modes and interrupts	Good understanding and explanation of instruction sets, addressing modes and interrupts	Excellent understanding and explanation of instruction sets, addressing modes and interrupts

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Explain the hierarchy of computer software	No or wrong explanation of computer software hierarchy	Limited and very basic explanation of computer software hierarchy	Basic but limited explanation of computer software hierarchy	Good and competent explanation of computer software hierarchy	Excellent and complete explanation of computer software hierarchy
Describe alternative computer systems	No or insufficient description of performance enhancing technologies	Limited and very basic description of performance enhancing technologies	Basic and partly satisfactory description of performance enhancing technologies	Good description of performance enhancing technologies	Excellent and complete description of performance enhancing technologies
Discuss the role of Data Communications and Networks in computer systems	No or very trivial discussion of data communications protocols and components	A very basic discussion of data communications protocols and components	Basic but credible discussion of data communications protocols and components	Good and wide-ranging discussion of data communications protocols and components	Excellent and complete discussion of data communications protocols and components

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